

BEST PRACTICE SERIES

Condition Monitoring –
Field Inspections

Application Maintenance Site Management Component Rebuild Safety MARC Management

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1.0 Introduction

This Best Practice deals with Condition Monitoring and Maintenance Management. The on time detection of potential problems is the key requisite to achieve the ultimate goal of "Repairing Before Failure".

The Field Inspection process includes three major activities:

1. Operator Inspection,
2. Field technical Inspection
3. Pre - PM technical Inspection

These three activities take place in the field and compliment each other to achieve the objectives defined below.

Goal: On time detection of potential machine problems or failures.

Objectives: Detect and register > 80% of pending visual defects of machines.

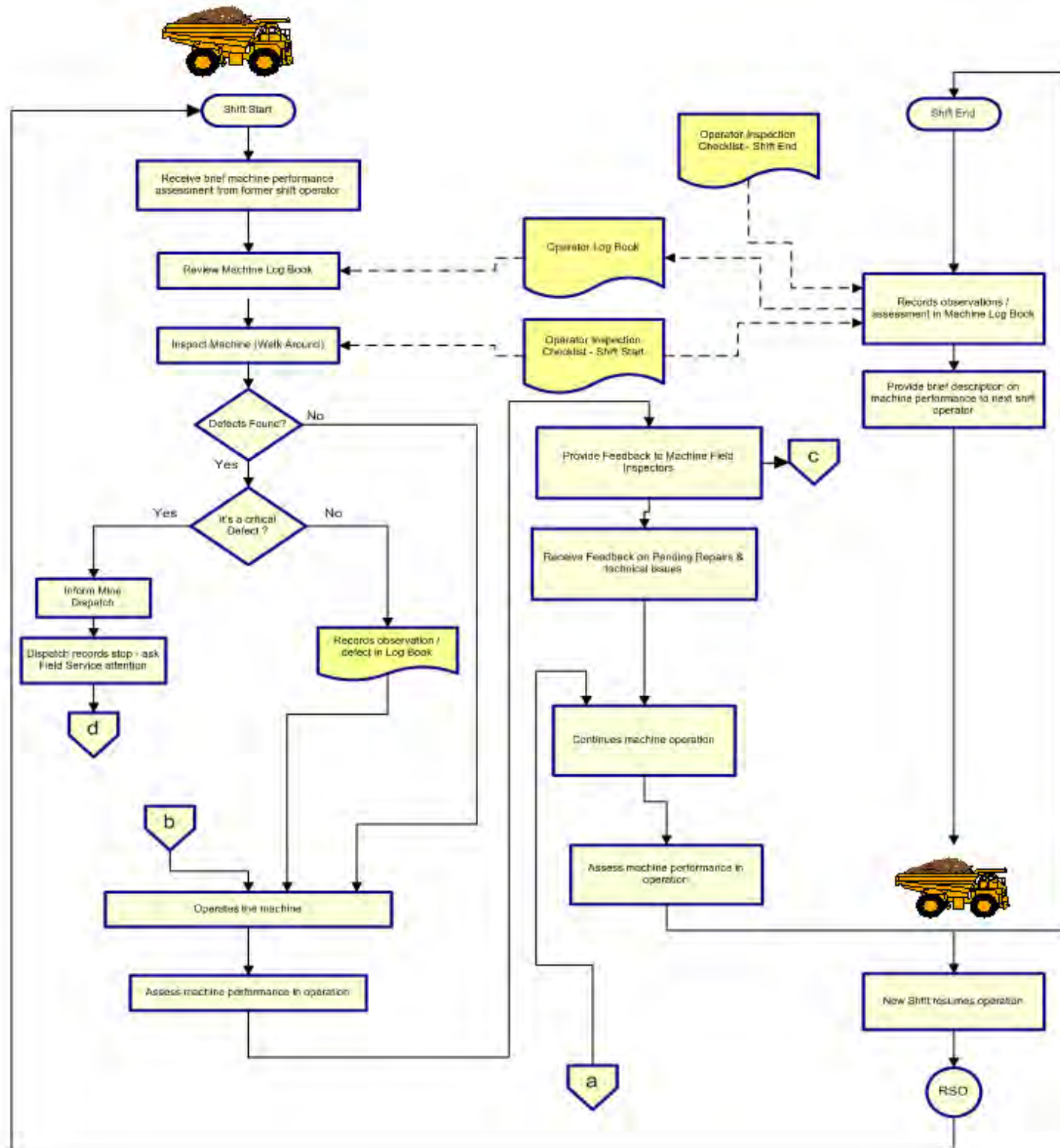
Scope: Detection of machine problems by visual inspections and basic predictive tests. It applies to all machines operated at the site.

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2.0 Best Practice Description

2.1 Field Inspection Process Overview (Operator Inspections)



The diagram above describes the recommended process for the Operator Inspections.

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The critical points to observe when utilizing this process are:

2.1.1 Log Book

A logbook should be implemented. The logbook must remain in the machine cab and be reviewed by the inspector at the time of the machine inspection.

Implementation of a logbook and the inspector interview of the operator to reduce to a minimum the need for double checks.

2.1.2 Inspector

Operator interview. The inspector should interview the operator when performing the machine inspection and understand his feedback, validate and transform it into a technical repair request (Backlog Request).

2.1.3 Operator Checklist

The operator check list included (printed) in the logbook should be structured into two major sections:

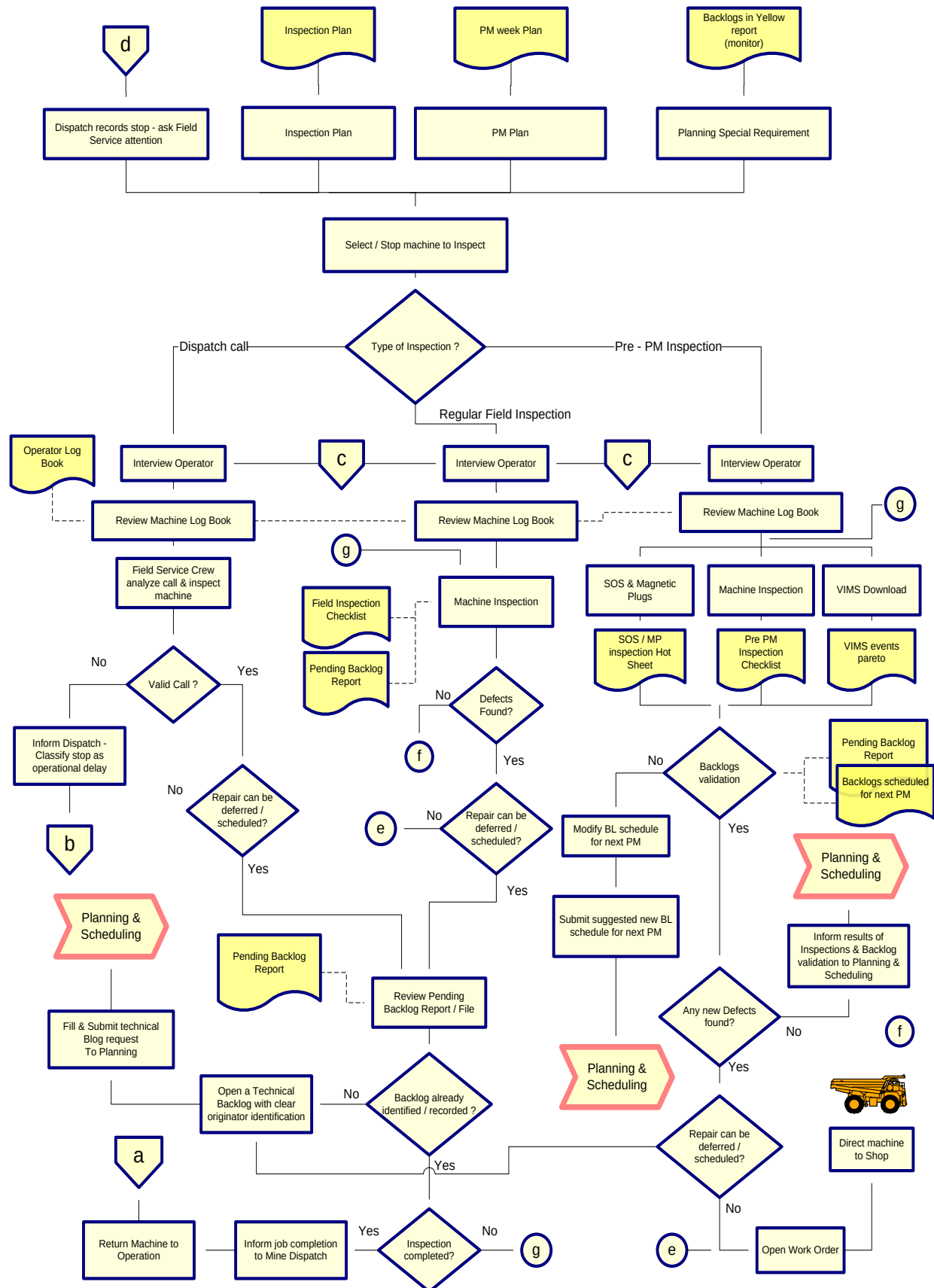
- **Shift start section:** To be completed through a visual inspection (operator walk around) prior to the start of the shift.
- **Shift end section:** Completed at the end or during the shift based on the operational assessment of the machine.

2.1.4 Operator Backlogs

The Backlogs that are generated as a result of operator feedback should identify clearly the name and / or ID of the operator to facilitate feedback and possible recognition.

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2.2 Field Inspection Process Overview (Technical Field Inspections)

The diagram on the previous page illustrates how the Technical Field Inspection Process should flow.

Critical points to observe when utilizing this process are:

2.2.1 Inspector-Operator Interview.

Technical Field Inspections should incorporate all routines from the operator interview and the analysis of the Operator Log Book.

2.2.2 Avoid Backlog Duplication

Before generating a Backlog the inspector must review the Backlog pending list report to avoid duplication of entries.

2.2.3 Pending Backlog Report.

The use of this report is also a big contributor to the efficiency and effectiveness of the inspections. The correct use of this report allows the inspectors to concentrate on the detection of new defects.

2.2.4 Measure the results

Measure the results /contribution of all the different inspection routines.

In order to assess the contribution of the various inspection routines, be prepared to classify the Backlogs generated in those routines.

3.0 Implementation Steps

1. Identify and train Field Inspectors.
2. Provide logbooks and training to the Operators.
3. Provide a designated location for field inspections and operator interviews. Some mines have used the fuel island and the refueling stop to perform the daily inspection and operator interview.
4. Develop site-specific inspection checklists for each model of machine.
5. Provide the inspectors with the current backlog list for each machine to be inspected.
6. Implement the Best Practice for Backlog Management.

4.0 Benefits

The on time detection of potential problems is the key requisite to achieve the ultimate goal of "Repairing Before Failure".

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The on time detection, registration (Backlog generation) and communication of potential problems allows Planning and Scheduling to:

1. Implement and manage an effective Backlog function.
2. Schedule the necessary repairs at the right time (At the right window of opportunity and Before Failure)
3. Provide effective use of all windows of opportunities (Machine stops).

4.1 Measurable Benefits

4.1.1 Time and Cost

The difference in time and cost between a repair executed before failure and an unscheduled one has been established in the range of 5 to 8.

4.1.2 Quality

MTBS as a reliability indicator is also positively influenced by a well planned and scheduled repair. Effectively using windows of opportunities is the first contributor to MTBS results. By performing scheduled repairs, the pressure from operations is reduced, allowing the service crew to better concentrate and executes the repairs more effectively.

5.0 Resources Required

A successful Field Inspection Program requires a minimum level of resources located in all areas indicated above.

5.1 Facilities:

A suitable location has to be implemented / designated in the field to perform the inspections. Some efficient mines have implement field facilities to perform the inspections in conjunction with other operational required activities, such as refueling and tire management. (See supporting attachments).

5.2 Tools:

The field crew has to be equipped with all inspection and predictive tools defined in the inspection checklists, such as infrared temperature gauge, pressure gauges, magnifying devices, penetrating inks, etc.

5.3 Support Equipment:

Equipped vehicles must be provided to assist the inspections routines. There will be cases in which the machines will have to be inspected in another location of the mine. (See supporting attachments).

5.4 Labor:

Trained Inspectors and Operators are key for the success of the inspections. Operators must receive basic technical training and inspectors must be trained in specialized predictive techniques.

Investment will depend on the specific site conditions.

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6.0 Supporting Attachments



[Field Inspection Tips & Examples](#) - PowerPoint Presentation (Click on “Attachments” tab within this document to view).

7.0 Related Best Practices

0806-2.70-1008 -Backlog Management

8.0 Acknowledgements

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